

Columbia Engineering Student
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EDUCATION

Columbia University, School of Engineering and Applied Science (SEAS), New York, NY
Bachelor of Science, Computer Engineering Expected May 2027

Washington & Jefferson College, Washington, PA
Bachelor of Arts, Computing and Information Sciences May 2025

Relevant Coursework: Computer Simulations, Physics, Differential Equations, Data Structures and Algorithms (Java), Advanced Programming (C), Probability for Engineers, NLP, Fundamentals of Computer Systems

HONORS AND GRANTS

Roy G. Post Foundation Undergraduate Scholar (2026)
College Sports Communicators (CSC) Academic All-District At Large (2024)
Association of Collegiate Water Polo Coaches, All-Academic (Outstanding) (2023, 2024)

RESEARCH EXPERIENCE

Pacific Northwest National Laboratory, Radiological Materials Team

Department of Energy Technical Intern July 2025-Present

Advisors: Jacob Anderson, Matt Asmussen

Collaborated with multidisciplinary teams at PNNL to modernize data management and research accessibility for nuclear waste materials studies. Co-developed a public-facing grout database website that functions as both a data entry portal and research tool, enabling external researchers to contribute data, access curated datasets, and generate custom visualizations to support analysis. In parallel, partnered with a Radiological Materials researcher to design an automated digital workflow for EPA 1315 leaching test data, integrating an Excel-based system with a custom script to produce standardized, publication-ready results and replace error-prone manual processes. Together, these tools improve data accuracy, consistency, and accessibility while reducing processing time and fostering collaboration across the nuclear waste research community.

Pacific Northwest National Laboratory, Radiological Materials Team

Department of Energy's Scientific Undergraduate Laboratory Internship program (SULI) May 2025 – July 2025

Advisors: Jacob Anderson, Matt Asmussen

Contributed to the development of computational models used to predict the long-term behavior of materials for nuclear waste management. Conducted in-depth analysis of academic research literature to extract, organize, and log relevant data into a research database, improving accessibility and usability for researchers. Communicated research findings through a poster presentation at a symposium alongside peers and other researchers.

Columbia University, Department of Physics

Lab Assistant September 2025-Present

Advisor: Jocelyn Rosenfeld

Set up and maintain laboratory equipment for undergraduate physics classes, ensure all apparatuses are organized and functional, and assist in resolving technical or equipment issues as they arise. Help in creation of lab manuals ensuring that directions are clear and allow the experiments to be completed by all students

PRESENTATIONS

Summer Research Symposium, Richland, WA, July 2026. Spicer, K. "Enhancing Research Workflow with Python-Based Automation for Data Management" (Poster).

Waste Management Symposia, Phoenix, AZ, March 2026. Spicer, "Enhancing Research Workflow with Python-Based Automation for Data Management" (Poster)

PROJECTS

C Based Web Server (C)

Designed and implemented a custom HTTP/1.0 web server in C supporting GET requests, with robust request parsing, standards-compliant status code handling, and secure URI validation. Enabled serving of static web pages and dynamic content by integrating with a custom database backend, forming a three-tier architecture (presentation, application, and data layers). Implemented directory handling, detailed request logging, and fault tolerance against malformed requests and client-side failures.

Free Throw Simulation (Python/Flask, HTML, CSS)

Led a seven-member team in developing a physics-based basketball free-throw simulation inspired by academic literature. Designed and implemented the core projectile motion model incorporating drag and Magnus forces and integrated the Python simulation with an HTML interface using Flask. Delivered both technical and non-technical demonstrations

Bike Sharing Predictor (Java/JavaFX)

Created a predictive model using kNN (k-Nearest Neighbors) algorithm to forecast the number of bike rentals for the upcoming hour based on historical bike share data. The model's accuracy was evaluated by predicting rental counts for a separate test set, and we achieved impressive results with minimal prediction errors.

AI Integration in Academic Affairs

Collaborated with a 15-member cohort on a semester-long systems analysis examining the implementation of AI in academic affairs using a design thinking framework. Conducted stakeholder interviews to identify key challenges and developed AI-driven solutions to personalize student support. Produced a comprehensive design brief detailing the proposed implementation strategy, including user stories, data flow diagrams, and functional prototypes.

LEADERSHIP/SERVICE

Washington & Jefferson Varsity Water Polo Team

Captain, September 2024 – May 2025

developed and implemented a dryland training schedule, organizing and leading captain's practices, and serving as the primary liaison between the team, coaching staff, and game officials.

Columbia University Club Water Polo

Secretary, September 2025 – Present

manage communications between the club and university administration to ensure efficient club coordination and operations, including necessary processes for our hosting of the Club Water Polo Championship in mid-November.

TECHNICAL SKILLS

Computational: Java/JavaFX, Python including Pandas, Pytorch, Flask, and Matplotlib, SQL, NoSQL, Tableau, Git, HTML, CSS, VBA, C, C++

Skills: High-Performance Computing, Quality Assurance, Data Modeling, Analysis, and Visualization, Socket Programming (TCP/IP), Three-Tier Architecture, Simulations, Digital Twins, NLP